Jingxuan Li

+1 424-440-0024 jingxuanli2030@u.northwestern.edu Evanston, IL Candy26i

Education

PhD **Northwestern University**, Statistics and Data Science — Evanston, IL, Sep. 2025 – present

- GPA: 3.9/4.0
- Coursework: Measure-Theoretic Probability, Statistical Inference, Linear Models.

BS **University of California, Los Angeles**, Mathematics of Computation — Los Angeles, CA, Sep. 2021 – Jun. 2025

- GPA: 3.98/4.0
- Coursework: Machine Learning, Optimization, Numerical Methods, ODEs, Real Analysis, Complex Analysis, Mathematical Modeling, Computer Graphics.
- Honors: Oda Abe Scholarship; Dean's Honors List (2021–2024).

Publications

Quantifying Preferences of Vision-Language Models via Value Decomposition in Social Media Contexts
Jingxuan Li, Yuning Yang, Shengqi Yang, Linfan Zhang, Ying Nian Wu
ACL 2025 Main Conference

[In preparation] Draft-Conditioned Routing for Multi-Agent LLM Systems: Estimating the Marginal Value of an Additional Model Call
Target venue: ICLR 2027

Research Experience

Marginal Value of Model Calls in LLM Multi-Agent Systems — Evanston, IL, May 2025 – present
Advisor: Prof. Ying Nian Wu (UCLA)

- Built a **manager–advisor** LLM system in which a manager model routes among three specialist sub-agents (extractor, reasoner, verifier), replacing the usual fixed pipeline with a learned decision of *whether* and *which* advisor to call.
- Formulated a **draft-conditioned marginal-value** objective: the manager first commits its own draft answer, counterfactual branches force each advisor in turn, and the shortest branch that converts a wrong draft into a correct one becomes the supervision target – yielding per-advisor **rescue vs. corruption** rates and an oracle headroom of **+0.36** over direct answering on MedQA.
- Post-trained managers at **4B/8B/9B** scale with cold-start **SFT** distilled from a DeepSeek teacher followed by **GRPO**, evaluated on **MedQA, GPQA, LegalBench and MMLU-Pro**.
- Currently drafting the manuscript.

Latent and Induced Persona in Vision-Language Models — Los Angeles, CA, Sep. 2023 – May 2025
Advisor: Prof. Ying Nian Wu (UCLA)

- Curated a **vectorized image dataset** from social-media short-form content through an automated **GUI-driven capture pipeline**, producing a reusable benchmark of screenshots and metadata.
- Benchmarked the value preferences of **Claude 3.5**, **Gemini 1.5**, **GPT-4o**, **BLIP-2** and **CogVLM** against **Schwartz's ten value dimensions**, decomposing model sentiment into interpretable value axes.
- Designed a pipeline for **inducing personas** in VLMs and measuring their role-play fidelity by alignment with social-media recommendation behaviour.
- Developed the **Inductive Scoring Questionnaire (ISQ)** prompting strategy, improving the consistency of role-playing VLMs; work published at **ACL 2025**.

Internship Experience

Tencent (CSIG – Cloud & Smart Industries Group), Big Data Research Intern — Beijing, China
Aug. – Sep. 2024

- Researched geolocation optimization, applying **NLP** tools (**jieba**, **ckip-transformers**) to match WiFi SSIDs against point-of-interest names for precise store-level localization.
- **Fine-tuned HunYuan-7B** on a purpose-built dataset, improving matching accuracy over the rule-based baseline.
- Shipped the matching algorithm as a production pipeline inside **Tencent Maps**, raising geolocation accuracy in live industrial use.

Tencent (CSIG – Cloud & Smart Industries Group), Technical Research Intern — Beijing, China
Jul. – Sep. 2023

- Designed **mathematical consistency checks** to detect contradictions in apartment geospatial records, improving data reliability upstream of downstream models.
- Reduced the time and space complexity of the detection algorithm in **Python**, cutting runtime on the full record set.

Skills

Languages: Python, C++, JavaScript, MATLAB, SQL, R, LaTeX
ML & LLM: PyTorch, HuggingFace Transformers, PEFT/LoRA, TRL (GRPO), DeepSpeed, vLLM, TensorFlow, Keras, Scikit-learn
Data & Infra: NumPy, Pandas, Matplotlib, PostgreSQL, Milvus, Prisma, AWS, Linux, Git, Tableau